INCH-POUND
HH-P-151G
July 14, 2020
SUPERSEDING
HH-P-151F
July 19, 1985

FEDERAL SPECIFICATION

PACKING; RUBBER-SHEET, CLOTH-INSERT

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. Cloth-insert, rubber-sheet packing covered by this specification is intended for flange joints for water or brine services up to 250 pounds of pressure and for ventilating systems.

1.2 Classification.

1.2.1 Classes. Packing covered by this specification shall be of the following classes as specified (see 6.2):

a. Class 1 – Commercial.
b. Class 2 – Polychloroprene rubber.
c. Class 3 – Styrene-butadiene rubber.
d. Class 4 – Acrylonitrile-butadiene rubber.

1.2.2 Thickness. Packing covered by this specification shall be of the thickness specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

MILITARY SPECIFICATIONS

MIL-DTL-16884 - Fuel, Naval Distillate

MILITARY STANDARDS

MIL-STD-129 - Military Marking for Shipment and Storage

(Copies of these documents are available online at https://quicksearch.dla.mil or https://assist.dla.mil.)

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 5330

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 411-1 - Hazardous Material Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Private sector and civil agencies may purchase copies of this voluntary standard from https://webstore.ansi.org.)

ASTM INTERNATIONAL

ASTM D395	-	Standard Test Methods for Rubber Property – Compression Set
ASTM D412	-	Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers – Tension
ASTM D413	-	Standard Test Methods for Rubber Property – Adhesion to Flexible Substrate
ASTM D471	-	Standard Test Method for Rubber Property – Effect of Liquids
ASTM D573	-	Standard Test Method for Rubber – Deterioration in an Air Oven
ASTM D865	-	Standard Test Method for Rubber – Deterioration by Heating in Air (Test Tube Enclosure)
ASTM D1777	-	Standard Test Method for Thickness of Textile Materials
ASTM D3773/D3773M	-	Standard Test Methods for Length of Woven Fabric
ASTM D3774	-	Standard Test Method for Width of Textile Fabric
ASTM D3776/D3776M	-	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D3951	-	Standard Practice for Commercial Packaging
ASTM D5034	-	Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

(Private sector and civil agencies may purchase copies of these voluntary standards from www.astm.org.)

3. REQUIREMENTS

3.1 First article inspection. A sample shall be subjected to first article inspection as specified (see 4.4.1, 6.2, and 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

3.2 Material. The packing shall consist of alternate layers of rubber compound and cotton fabric. Fabric shall be completely immersed in elastomer so that each fiber is coated with rubber (see 4.4.2).

3.2.1 Fabric. Cotton fabric used as the cloth insertion shall be a chafer duck in accordance with 3.4. The fabric shall be evenly and firmly woven and as free from unsightly defects, dirt, knots, lumps, and irregularities of twist as is consistent with good manufacturing practices (see 4.4.1).

3.2.2 Rubber classes.

3.2.2.1 Class 1. The rubber compound used in the manufacture of class 1 packing shall be natural rubber, synthetic rubber (other than butyl), or a mixture of natural and synthetic rubber other than butyl.

3.2.2.2 Class 2. The rubber compound used in the manufacture of class 2 packing shall utilize polychloroprene as the base material.

3.2.2.3 Class 3. The rubber compound used in the manufacture of class 3 packing shall utilize a copolymer product of butadiene and styrene as the base material.

3.2.2.4 Class 4. The rubber compound used in the manufacture of class 4 packing shall utilize a copolymer product of butadiene and acrylonitrile as the base material.

3.3 Construction. Both faces of the packing shall be of rubber. The layers of rubber compound shall be of equal and uniform thickness. There shall be one ply of fabric in packings less than ⅛ inch thick. Packings ⅛ inch thick and over shall be made with one ply of fabric for each 1/16-inch thickness.

3.4 Fabric insert. The physical properties of the fabric insert shall be as specified in 3.4.1 and 3.4.2.

3.4.1 Weight. The weight of the fabric shall be 10 to 12 ounces per square yard (see 4.4.3.1).

3.4.2 Breaking strength. The breaking strength of the fabric shall be not less than 125 pounds when tested as specified in 4.4.3.2.

3.5 Rubber compound. The physical properties of the rubber compound shall be as specified in 3.5.1 through 3.5.5.2.

3.5.1 Tensile strength.

3.5.1.1 Before aging. The tensile strength of the rubber compound before aging shall be not less than 1,500 pounds per square inch (see 4.4.4.1.1).

3.5.1.2 After aging (accelerated aging). The tensile strength of the rubber compound after accelerated aging as specified in 4.4.4.1.2 shall be not less than 80 percent of the tensile strength of the unaged rubber compound.

3.5.2 Ultimate elongation. The ultimate elongation of the rubber compound before aging shall be not less than 250 percent (see 4.4.4.2).

3.5.3 Compression set. The compression set of the rubber compound determined by the method specified in 4.4.4.3 shall be not more than 25 percent.

3.5.4 Resistance to water.

3.5.4.1 Swelling. The increase in volume of the rubber compound in water shall not exceed 20 percent when tested as specified in 4.4.4.4.1.

3.5.4.2 Extraction. The amount of water-soluble matter in the rubber compound shall not exceed 0.5 percent when tested as specified in 4.4.4.4.2.

3.5.5 Resistance to oils.

3.5.5.1 Class 2, polychloroprene rubber. The increase in volume of the rubber compound in class 2 packing shall not exceed 100 percent when tested as specified in 4.4.4.5.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

3.5.5.2 Class 4, acrylonitrile-butadiene rubber. The increase in volume of the rubber compound in class 4 packing shall not exceed 40 percent when tested as specified in 4.4.4.5.2.

3.6 Adhesive bond. When tested as specified in 4.4.5, the bond strength between rubber layers and fabric insertions shall be not less than 15 pounds per inch of width.

3.7 Width and weight of rolls. Unless otherwise specified (see 6.2), packing shall be furnished in rolls 35 to 44 inches wide with trimmed edges and weighing not less than 100 pounds (see 4.4.6).

3.8 Thickness. The thickness of the packing at any point shall not differ from the specified thickness by more than the following amounts (see 4.4.6):

a. For thickness 3⁄32 inch or less: ±0.016 inch

b. For thickness over 3⁄32 inch: ±0.02 inch

3.9 Workmanship. The surface shall be smooth and free from pits, lumps, and other defects that may affect the serviceability of the packing (see 4.4.1).

3.10 Toxicity. When evaluated in accordance with 4.5, the gasket material shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.5 and 6.4).

3.11 Prohibited materials. The gasket material shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows (see table I):

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

TABLE I. First article and quality conformance inspections.

Characteristic	Requirement	Test/Examination	Test Standard
Material 1/	3.2	4.4.2	
Construction	3.3	4.4.1	
Fabric insert:			
Weight	3.4.1	4.4.3.1	ASTM D3776/D3776M
Breaking strength	3.4.2	4.4.3.2	ASTM D5034
Rubber compound:			
Tensile strength	3.5.1.1	4.4.4.1.1	ASTM D412
Accelerated aging	3.5.1.2	4.4.4.1.2	ASTM D573 or ASTM D865
Ultimate elongation	3.5.2	4.4.4.2	ASTM D412
Compression set	3.5.3	4.4.4.3	ASTM D395
Resistance to water:			
Swelling	3.5.4.1	4.4.4.4.1	ASTM D471
Extraction	3.5.4.2	4.4.4.4.2	ASTM D471
Resistance to oils:			
Class 2	3.5.5.1	4.4.4.5.1	ASTM D471
Class 4	3.5.5.2	4.4.4.5.2	ASTM D471
Adhesive bond	3.6	4.4.5	ASTM D413
Width and weight	3.7	4.4.6	
Thickness	3.8	4.4.6	
Workmanship	3.9	4.4.1	
Toxicity	3.10	4.5	
Prohibited materials	3.11	4.6	

NOTE:
1/ First article inspection only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

4.2.1 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the conditions specified herein.

4.2.2 Production equipment inspection. An inspection shall be performed prior to starting a product run to ensure that equipment and process lines have been purged of material from the previous production run. The inspection shall be performed on all equipment used in manufacturing operations where the material is not segregated and positively identified.

4.3 Sampling.

4.3.1 Lot. For the purpose of sampling, a lot shall consist of all packing of the same type and thickness produced under essentially the same conditions.

4.3.2 Sampling for examinations and tests. A random sample shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II. The Acceptance Quality Limit (AQL) shall be 2.5 percent defective or as specified (see 6.2) for first article inspection and 4.0 percent defective or as specified (see 6.2) for quality conformance inspection. Each sample piece shall be 12 by 12 inches or 3 feet long by ordered width. Unless otherwise specified (see 6.2), a piece of fabric 12 by 24 inches taken from the lot of fabric used in the manufacture of the packing delivered shall be furnished with each delivery of packing.

4.4 Inspections. The following examinations and tests apply to both first article and quality conformance inspections, except where specified (see table I).

4.4.1 Visual and dimensional examination. Each of the samples shall be subjected to surface examinations for construction, identification, workmanship, dimensions, and tolerances; as well as measured and weighed, to determine conformance with the requirements of this specification that do not require tests. Any roll in the sample containing one or more visual or dimensional defects shall not be offered for delivery, and if the number of defective rolls in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.4.2 Material examination (first article inspection only). Material examination shall consist of certification supported by verifying data that the material specified in 3.2, 3.2.1, and 3.2.2 is in conformance with the referenced specification or requirements.

4.4.3 Fabric. If the sample of fabric fails to meet either of the requirements for which tests were made, the lot of packing represented by the sample shall be considered as having failed to meet the requirements of this specification.

4.4.3.1 Weight. The sample of fabric shall be tested for weight as specified in ASTM D3776/D3776M (see 3.4.1).

4.4.3.2 Breaking strength. The breaking strength of the fabric in the packing shall be determined as specified in ASTM D5034 using the modified grab, as applicable (see 3.4.2).

4.4.4 Rubber compound.

4.4.4.1 Tensile strength.

4.4.4.1.1 Before aging. The tensile strength of the rubber compound in the packing shall be determined as specified in ASTM D412. Die C or E shall be used for cutting the specimen (see 3.5.1.1).

4.4.4.1.2 After aging (accelerated aging). The rubber compound shall be exposed to accelerated aging as specified in ASTM D573 or ASTM D865 (see 3.5.1.2) except that:

a. Classes 1 and 3 shall be exposed at a temperature of 158±3.6 degrees Fahrenheit (°F) (70±2 degrees Celsius [°C]) for 70 hours.

b. Classes 2 and 4 shall be exposed at a temperature of 212±3.6 °F (100±2 °C) for 70 hours.

After accelerated aging, the compound shall be retested for tensile strength. The change in tensile strength shall be used to determine the amount of deterioration due to aging (see 3.5.1.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

4.4.4.2 Ultimate elongation. The ultimate elongation of the rubber compound in the packing shall be determined as specified in ASTM D412. Die C or E shall be used for cutting the specimen (see 3.5.2).

4.4.4.3 Compression set. The compression set of the rubber compound in the packing shall be determined as specified in ASTM D395, method B (see 3.5.3) except that:

a. The thickness of the specimen shall be 0.25±0.005 inch.

b. The limitation of 7 plies does not apply.

c. Classes 1 and 3 packing shall be exposed at a temperature of 158±3.6 °F (70±2 °C) for 22 hours.

d. Classes 2 and 4 shall be exposed at a temperature of 212±3.6 °F (100±2 °C) for 22 hours.

4.4.4.4 Resistance to water.

4.4.4.4.1 Swelling. Resistance to water shall be measured at 73.4±6 °F (23±-14.44 °C) for 46 hours using the change in volume (after immersion) test method as specified in ASTM D471 (see 3.5.4.1).

4.4.4.4.2 Extraction. Determine the amount of material extracted by immersion in boiling water using the procedure for change in mass as specified in ASTM D471 with boiling (212±4 °F [100±-15.56 °C]) distilled water used as the immersion liquid for a duration of 1 hour (see 3.5.4.2).

4.4.4.5 Resistance to oils. (see 3.5.5)

4.4.4.5.1 Class 2, polychloroprene rubber. The change in volume of the rubber compound in class 2 packing shall be measured using the change in volume (after immersion) test method as specified in ASTM D471 with MIL-DTL-16884 (F-76) used as the immersion liquid for a duration of 70±¼ hours at a temperature of 212±3.6 °F (100±2 °C), to determine swelling (see 3.5.5.1).

4.4.4.5.2 Class 4, acrylonitrile-butadiene rubber. The change in volume of the rubber compound in class 4 packing shall be determined as specified in 4.4.4.5.1 (see 3.5.5.2).

4.4.5 Adhesive bond. The bond strength between rubber layers and fabrics shall be determined as specified in ASTM D413 (see 3.6).

4.4.6 Physical dimensions and weight of rolls. The physical dimensions and weight shall be as specified (see 3.7 and 3.8).

4.5 Toxicity. A Health Hazzard Assessment (HHA) will be conducted to ensure conformance to 3.10 as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the gasket material using data provided by the manufacturer/distributor to the NMCPHC (see 3.10 and 6.4).

4.6 Prohibited materials. Prohibited materials shall be verified as specified (see 3.11 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Packing and marking. Commercial packing should be in accordance with ASTM D3951. In addition to any special marking required (see 6.2), interior packs and exterior shipping containers are to be marked in accordance with MIL-STD-129 for levels A, B, and C and include bar code marking. Commercial interior packs and shipping containers are to be marked in accordance with ASTM D3951 and also include bar code marking as specified in MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

6. NOTES

INFORMATION FOR GUIDANCE ONLY. (This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. Packing covered by this specification is intended for use as follows:

a. Class 1 – When the class of rubber and resistance to oil are not important, the maximum use temperature is a nominal 200 °F (93.33 °C).

b. Class 2 – When medium resistance to swelling in petroleum base hydrocarbons is desired, the maximum use temperature is a nominal 200 °F (93.33 °C).

c. Class 3 – When synthetic rubber is desired and resistance to swelling in petroleum base hydrocarbons is not essential, the maximum use temperature is a nominal 200 °F (93.33 °C).

d. Class 4 – When high resistance to swelling in petroleum base hydrocarbons is desired, the maximum use temperature is a nominal 250 °F (121.11 °C).

This product must survive in a maritime environment in excess of commercial standards and operate at higher levels of reliability to meet the Navy's mission.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity of packing desired in square yards.

c. Class and thickness required (see 1.2).

d. First article sample, if required (see 3.1).

e. Width and weight of rolls, if other than indicated (see 3.7).

f. Requirements for an HHA (see 3.10 and 4.5).

g. Requirement for prohibited materials verification (see 3.11 and 4.6).

h. Inspection conditions, if other than indicated (see 4.2.1).

i. AQL requirements, if other than indicated (see 4.3.2).

j. Requirements for the piece of fabric sample taken from the lot of fabric used in the manufacture of the packing furnished with each delivery, if other than indicated (see 4.3.2).

k. Packaging requirements (see 5.1).

l. Special marking required (see 5.2).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. A sample will be subjected to first article inspection if:

a. The manufacturer has not previously furnished the product to the Government.

b. The manufacturer previously furnished the product to the Government, but:

(1) There have been subsequent changes in processes or specifications,

(2) Production has been discontinued for an extended period of time, or

(3) The product acquired under a previous contract developed a problem during its life.

6.4 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrialhygiene/Pages/health-hazard-assessment.aspx.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

6.5 Subject term (key word) listing.

Acrylonitrile-butadiene rubber
Brine service
Flange joint
Gasket
Polychloroprene rubber
Styrene-butadiene rubber
Ventilating system
Water service

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

MILITARY INTERESTS:

Custodians:
Army – AR
Navy – SH
Air Force – 71

Preparing Activity:
Navy – SH
(Project 5330-2020-002)

Review Activities:
Army – AT, CR4
DLA – IS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.